Complex Analysis Homework 4

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2. Let u be a harmonic function in the unit disc that is continuous on its closure. Deduce Poisson's integral formula

$$u(z_0) = \frac{1}{2\pi}\int_0^{2\pi} \frac{1-|z_0|^2}{|e^{i\theta}-z_0|^2} u(e^{i\theta})d\theta \quad \text{for } |z_0|<1$$

from the special case $z_0 = 0$ (the mean value theorem). Show that if $z_0 = re^{i\varphi}$, then

$$\frac{1-|z_0|^2}{|e^{i\theta}-z_0|^2} = \frac{1-r^2}{1-2r\cos(\theta-\varphi)+r^2} = P_r(\theta-\varphi),$$

and we recover the expression for the Poisson kernel derived in the exercises of the previous chapter.

[Hint: Set $u_0(z) = u(T(z))$ where

$$T(z) = \frac{z_0 - z}{1-\overline{z_0}z}.$$

Prove that u_0 is harmonic. Then apply the mean value theorem to u_0, and make a change of variables in the integral.]

证明 根据第八章引理 *1.3* 可知 u_0 也是调和的

$$u_0(0) = u(z_0) = \frac{1}{2\pi}\int_0^{2\pi} u_0(e^{i\varphi})d\varphi.$$

作换元 $T(e^{i\varphi}) = e^{i\theta}$, 即有

$$e^{i\varphi} = T(e^{i\theta}) = \frac{z_0 - e^{i\theta}}{1-\overline{z_0}e^{i\theta}},$$

两边求导

$$d\varphi = \frac{(|z_0|^2-1)e^{i\theta}}{(1-\overline{z_0}e^{i\theta})^2}e^{-i\varphi}d\theta = \frac{(|z_0|^2-1)e^{i\theta}}{(1-\overline{z_0}e^{i\theta})(z_0-e^{i\theta})}d\theta$$
$$= \frac{(|z_0|^2-1)}{(e^{-i\theta}-\overline{z_0})(z_0-e^{i\theta})}d\theta = \frac{1-|z_0|^2}{|e^{i\theta}-z_0|^2}d\theta,$$

代入即得

$$u(z_0) = \frac{1}{2\pi}\int_0^{2\pi}\frac{1-|z_0|^2}{|e^{i\theta}-z_0|^2}u(e^{i\theta})d\theta.$$

1. A holomorphic mapping $f : U \to V$ is a local bijection on U if for every $z \in U$ there exists an open disc $D \subset U$ centered at z, so that $f : D \to f(D)$ is a bijection.

Prove that a holomorphic map $f : U \to V$ is a local bijection on U if and only if $f'(z) \neq 0$ for all $z \in U$.

[Hint: Use Rouché's theorem as in the proof of Proposition 1.1.]

证明 必要性：根据命题 *1.1* 即得

充分性：若 $f'(z_0) \neq 0$, 不妨设 $f(z_0) = w_0$, 从而存在 $B(z_0, \delta)$ 使得 $f(z) \neq f(z_0), z \in B(z_0, \delta)$ 且 $z \neq z_0$, 任取 $|w| < w_m = inf_{|z-z_0|=\delta}|f(z)-w_0|$, 于是考虑 $g(z) = f(z) - w$, 根据 *Rouché's theorem* 有 $g(z)$ 在 $B(z_0, \delta)$ 上只有一个零点, 从而 f 是从 $B(z_0, \delta)$ 到 $f(B(z_0, \delta)) = B(w_0, w_m)$ 的双射

2. Suppose $F(z)$ is holomorphic near $z = z_0$ and $F(z_0) = F'(z_0) = 0$, while $F''(z_0) \neq 0$. Show that there are two curves Γ_1 and Γ_2 that pass through z_0, are orthogonal at z_0, and so that F restricted to Γ_1 is real and has a minimum at z_0, while F restricted to Γ_2 is also real but has a maximum at z_0.

[Hint: Write $F(z) = (g(z))^2$ for z near z_0, and consider the mapping $z \mapsto g(z)$ and its inverse.]

证明 设 g 使得在 z_0 的邻域上有 $F(z) = (g(z))^2$, 从而可设

$$g(z) = a(z - z_0) + \cdots$$

根据前一题结论, 存在 δ 使得 g 在 $B(0, \delta)$ 上是单射且与 $B(0, w_m)$ 是共形

考虑 g^{-1}, 在实轴和虚轴下的像, 分别设为 Γ_1, Γ_2, 于是在 Γ_1 上 $f \geq 0$, 在 Γ_2 上 $f \leq 0$, 且极值点恰好在 $z = z_0$ 处取到。并且根据共形的性质, 这两条曲线在 z_0 处仍然正交

3. Suppose U and V are conformally equivalent. Prove that if U is simply connected, then so is V. Note that this conclusion remains valid if we merely assume that there exists a continuous bijection between U and V.

证明 由于 U 和 V 是共形的，所以存在全纯的双射 $f : U \to V$

任取 V 中一条闭曲线 $\gamma\colon [0,1] \to V$，令

$$\alpha = f^{-1} \circ \gamma\colon [0,1] \to U.$$

则 α 是 U 中的一条闭曲线。

因为 U 单连通，α 可在 U 中连续收缩到一点，即存在连续映射

$$H\colon [0,1] \times [0,1] \to U$$

满足

$$H(t,0) = \alpha(t), \quad H(t,1) = z_0,$$
$$H(0,s) = H(1,s) = z_0,$$

其中 z_0 为常数点。

令 $\widetilde{H}(t,s) = f(H(t,s))$，则 $\widetilde{H}$ 连续，且

$$\widetilde{H}(t,0) = \gamma(t), \quad \widetilde{H}(t,1) = f(z_0),$$
$$\widetilde{H}(0,s) = \widetilde{H}(1,s) = f(z_0).$$

因此 γ 可在 V 中收缩为一点。由 γ 的任意性，V 单连通。

4. Does there exist a holomorphic surjection from the unit disc to $\mathbb{C}$?

[Hint: Move the upper half-plane "down" and then square it to get $\mathbb{C}$.]

证明 考虑 $f = (i\dfrac{1-z}{1+z} - i)^2$, 即将单位圆盘映射到上半平面，向下平移后再平方。

5. Prove that $f(z) = -\frac{1}{2}\left(z + \frac{1}{z}\right)$ is a conformal map from the half-disc $\{z = x + iy : |z| < 1,\ y > 0\}$ to the upper half-plane.

[Hint: The equation $f(z) = w$ reduces to the quadratic equation $z^2 + 2wz + 1 = 0$, which has two distinct roots in $\mathbb{C}$ whenever $w \neq \pm 1$. This is certainly the case if $w \in \mathbb{H}$.]

证明 设 $z = x + iy$, 于是

$$x + iy + \frac{1}{x+iy} = x + \frac{x}{x^2+y^2} + i(y - \frac{y}{x^2+y^2})$$

$y - \dfrac{y}{x^2+y^2} < 0$，于是像在上半平面中

对于上半平面中的任何一点 w，不妨设 $f(z) = w$，从而

$$z^2 + 2zw + 1 = 0$$

于是对于两个根有

$$z_1 z_2 = 1, z_1 + z_2 = -2w$$

不妨设 $|z_1| < 1 < |z_2|$, 于是 $Imz_1 + Imz_2 = Imz_1(1 - \dfrac{1}{|z_1|^2}) < 0$, 从而 $Imz_1 > 0$，即 z_1 属于上半圆盘

而 $f'(z) = -\dfrac{1}{2}(1 - \dfrac{1}{z^2}) \neq 0$，综上即有 *$f$is a conformal map from the half-disc* $\{z = x + iy : |z| < 1,\ y > 0\}$ *to the upper half-plane.*

9. Prove that the function u defined by

$$u(x, y) = Re\left(\frac{i+z}{i-z}\right) \quad \text{and} \quad u(0, 1) = 0$$

is harmonic in the unit disc and vanishes on its boundary. Note that u is not bounded in $\mathbb{D}$.

证明 令 $f(z) = \dfrac{i+z}{i-z}$。在单位圆盘 $\mathbb{D} = \{z : |z| < 1\}$ 内，分母 $i - z \neq 0$, $f(z)$ 是 $\mathbb{D}$ 内的全纯函数。因此 $u = \Re(f(z))$ 在 $\mathbb{D}$ 内调和。

设 $|z| = 1$，则 $z\overline{z} = 1$，即 $\overline{z} = \dfrac{1}{z}$。

$$\frac{i+z}{i-z} = \frac{(i+z)\overline{(i-z)}}{|i-z|^2} = \frac{(i+z)(-i-\overline{z})}{|i-z|^2}$$

计算分子:

$$(i+z)(-i-\overline{z}) = 1 - i\overline{z} + iz - z\overline{z} = i(z - \overline{z})$$

该结果实部为 *0*，即:

$$u(x, y) = \Re\left(\frac{i+z}{i-z}\right) = 0, \quad \forall |z| = 1.$$

取沿虚轴逼近边界的点列: $z = iy$，其中 $0 < y < 1$。

$$u(0, y) = \Re\left(\frac{i+iy}{i-iy}\right) = \Re\left(\frac{i(1+y)}{i(1-y)}\right) = \Re\left(\frac{1+y}{1-y}\right) = \frac{1+y}{1-y}$$

当 $y \to 1^-$ 时，

$$u(0,y) = \frac{1+y}{1-y} \to +\infty$$

故函数 $u(x,y)$ 在单位圆盘 $\mathbb{D}$ 内无界。